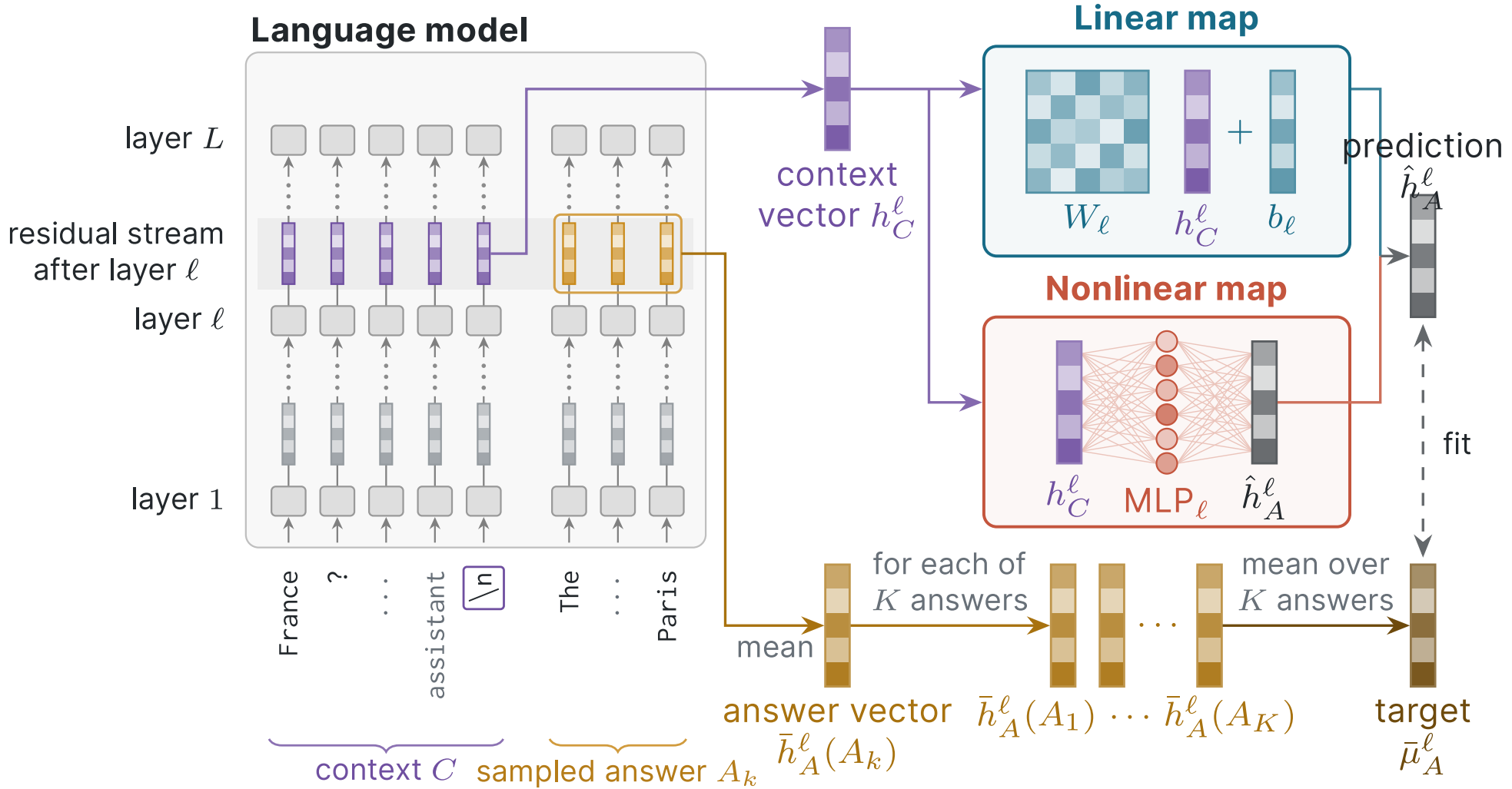

Language model
layer L
residual stream after layer ℓ
layer ℓ
layer 1
France
?
...
assistant
\n
The
...
Paris
context C
sampled answer A_k
context vector h_C^ℓ
Linear map
W_ℓ
h_C^ℓ
b_ℓ
prediction $\hat{h}_A^\ell$
Nonlinear map
h_C^ℓ
MLP_ℓ
$\hat{h}_A^\ell$
fit
mean
answer vector $\bar{h}_A^\ell(A_k)$
for each of K answers
$\bar{h}_A^\ell(A_1)$ ... $\bar{h}_A^\ell(A_K)$
mean over K answers
target $\bar{\mu}_A^\ell$